

Table 7-63. Get DCD Info Response Payload (Sheet 2 of 2)

Byte Offset	Length in Bytes	Description
14h	8	Region 0 Supported Block Size Mask: Indicates the block sizes that the region supports. Each bit indicates a power of 2 supported block size, where bit n being set indicates that block size 2^n is supported. Bits[5:0] and bits[63:52] shall be 0. At least one block size shall be supported.
1Ch	8	Region 1 Supported Block Size Mask: As defined in Region 0 Supported Block Size Mask. Valid only if Number of Supported Regions > 1.
24h	8	Region 2 Supported Block Size Mask: As defined in Region 0 Supported Block Size Mask. Valid only if Number of Supported Regions > 2.
2Ch	8	Region 3 Supported Block Size Mask: As defined in Region 0 Supported Block Size Mask. Valid only if Number of Supported Regions > 3.
34h	8	Region 4 Supported Block Size Mask: As defined in Region 0 Supported Block Size Mask. Valid only if Number of Supported Regions > 4.
3Ch	8	Region 5 Supported Block Size Mask: As defined in Region 0 Supported Block Size Mask. Valid only if Number of Supported Regions > 5.
44h	8	Region 6 Supported Block Size Mask: As defined in Region 0 Supported Block Size Mask. Valid only if Number of Supported Regions > 6.
4Ch	8	Region 7 Supported Block Size Mask: As defined in Region 0 Supported Block Size Mask. Valid only if Number of Supported Regions > 7.

7.6.7.6.2 Get Host DC Region Configuration (Opcode 5601h)

This command retrieves the Dynamic Capacity configuration for an LD-FAM DCD, for a specified host.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-64. Get Host DC Region Configuration Request Payload

Byte Offset	Length in Bytes	Description
0h	2	Host ID: For an LD-FAM device, the LD-ID of the host interface configuration to query.
2h	1	Region Count: The maximum number of region configurations to return in the output payload.
3h	1	Starting Region Index: Index of the first requested region.

Table 7-65. Get Host DC Region Configuration Response Payload

Byte Offset	Length in Bytes	Description
0h	2	Host ID: For an LD-FAM device, the LD-ID of the host interface configuration returned.
2h	1	Number of Available Regions: As defined in Get Dynamic Capacity Configuration Output Payload.
3h	1	Number of Regions Returned: The number of entries in the Region Configuration List.
4h	Varies	Region Configuration List: DC Region Info for region specified via Starting Region Index input field. The format of each entry is defined in Table 7-66.
Varies	4	Total Number of Supported Extents: Total number of extents that the device supports on this LD.
Varies	4	Number of Available Extents: Remaining number of extents that the device supports, as defined in Section 9.13.3.3.
Varies	4	Total Number of Supported Tags: Total number of Tag values that the device supports on this LD.
Varies	4	Number of Available Tags: Remaining number of Tag values that the device supports, as defined in Section 9.13.3.3.

Table 7-66. DC Region Configuration

Byte Offset	Length in Bytes	Description
00h	8	Region Base: As defined in Table 8-180.
08h	8	Region Decode Length: As defined in Table 8-180.
10h	8	Region Length: As defined in Table 8-180.
18h	8	Region Block Size: As defined in Table 8-180.
20h	1	Note: More than one bit may be set at a time. - Bits[1:0]: Reserved - Bit[2]: NonVolatile: As defined in the Flags field of Device Scoped Memory Affinity Structure defined in Coherent Device Attribute Table (CDAT) Specification - Bit[3]: Sharable: As defined in the Flags field of Device Scoped Memory Affinity Structure defined in CDAT Specification - Bit[4]: Hardware Managed Coherency: As defined in the Flags field of Device Scoped Memory Affinity Structure defined in CDAT Specification - Bit[5]: Interconnect specific Dynamic Capacity Management: As defined in the Flags field of Device Scoped Memory Affinity Structure defined in CDAT Specification - Bit[6]: Read-Only: As defined in the Flags field of Device Scoped Memory Affinity Structure defined in CDAT Specification - Bit[7]: Reserved
21h	3	Reserved
24h	1	- Bit[0]: Sanitize on Release: As defined in Table 8-180 - Bits[7:1]: Reserved
25h	3	Reserved

7.6.7.6.3 Set DC Region Configuration (Opcode 5602h)

This command sets the configuration of a DC Region for an LD-FAM DCD. This command shall be processed only when all capacity has been released from the region on all LDs. The device shall generate an Event Record of type Region Configuration Updated upon successful processing of this command.

This command shall fail with Unsupported under the following conditions:

- When all capacity has been released from the DC Region on all hosts, and one or more blocks are allocated to the specified region
- When the Sanitize on Release field does not match the region's configuration, as reported from the Get Host DC Region Configuration, and the device does not support reconfiguration of the Sanitize on Release setting, as advertised by the Sanitize on Release Configuration Support Mask in the Get DCD Info response payload

This command shall fail with Invalid Security State under the following condition:

- In support of confidential computing, if the device has been locked while utilizing secure CXL TSP interfaces, the device shall reject any attempts to change the DCD configuration by returning Invalid Security State status. See [Section 11.5](#) for details on locking a device and locked device behavior.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required
- Invalid Security State

Command Effects:

- Configuration Change after Cold Reset
- Configuration Change after Conventional Reset
- Configuration Change after CXL Reset
- Immediate Configuration Change
- Immediate Data Change

Table 7-67. Set DC Region Configuration Request and Response Payload

Byte Offset	Length in Bytes	Description
0h	1	Region ID: Specifies which region to configure. Valid range is from 0 to 7.
1h	3	Reserved
4h	8	Region Block Size: As defined in Table 8-180 .
Ch	1	• Bit[0]: Sanitize on Release: As defined in Table 8-180 • Bits[7:1]: Reserved
Dh	3	Reserved

7.6.7.6.4 Get DC Region Extent Lists (Opcode 5603h)

This command sets the Dynamic Capacity Extent List for an LD-FAM DCD, for a specified host.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input

- Internal Error
- Retry Required

Command Effects:

- None

Table 7-68. Get DC Region Extent Lists Request Payload

Byte Offset	Length in Bytes	Description
0h	2	Host ID: For an LD-FAM device, the LD-ID of the host interface.
2h	2	Reserved
4h	4	Extent Count: The maximum number of extents to return in the output response. The device may not return more extents than requested; however, it can return fewer extents. 0 is valid and allows the FM to retrieve the Total Extent Count and Extent List Generation Number without retrieving any extent data.
8h	4	Starting Extent Index: Index of the first requested extent. A value of 0 will retrieve the first extent in the list.

Table 7-69. Get DC Region Extent Lists Response Payload

Byte Offset	Length in Bytes	Description
00h	2	Host ID: For an LD-FAM device, the LD-ID of the host interface query.
02h	2	Reserved
04h	4	Starting Extent Index: Index of the first extent in the list.
08h	4	Returned Extent Count: The number of extents returned in Extent List[].
0Ch	4	Total Extent Count: The total number of extents in the list.
10h	4	Extent List Generation Number: A device-generated value that is used to indicate that the list has changed.
14h	4	Reserved
18h	Varies	Extent List[]: Extent list for the specified host as defined in Table 8-63.

7.6.7.6.5 Initiate Dynamic Capacity Add (Opcode 5604h)

This command initiates the addition of Dynamic Capacity for an LD-FAM DCD, to the specified region on a host. This command shall complete when the device initiates the Add Capacity procedure, as defined in Section 8.2.10.2.2. The processing of the actions initiated in response to this command may or may not result in a new entry or multiple entries grouped via the More flag (see Table 8-62) in the Dynamic Capacity Event Log. To perform Dynamic Capacity Add on a GFD, see Section 8.2.10.9.10.7.

A Selection Policy is specified to govern the device's selection of which memory resources to add:

- **Free:** Unassigned extents are selected by the device, with no requirement for contiguous blocks
- **Contiguous:** Unassigned extents are selected by the device and shall be contiguous
- **Prescriptive:** Extent list of capacity to assign is included in the request payload
- **Enable Shared Access:** Enable access to extent(s) previously added to another host in a DC Region that reports the "Sharable" flag, as designated by the specified tag value

See [Section 9.13.3.2](#) for examples of how this command may be used to set up different types of sharing arrangements.

The command shall fail with Invalid Input under the following conditions:

- When the command is sent with an invalid Host ID, or an invalid region number, or an unsupported Selection Policy
- When the Length field is not a multiple of the Block size and the Selection Policy is either Free or Contiguous

The command, with selection policy Enable Shared Access, shall also fail with Invalid Input under the following conditions:

- When the specified region is not Sharable
- When the tagged capacity is already mapped to any Host ID via a non-Sharable region
- When the tagged capacity cannot be added to the requested region due to device-imposed restrictions
- When the same tagged capacity is currently accessible by the same LD

The command shall fail with Resources Exhausted when the length of the added capacity plus the current capacity present in all extents associated with the specified region exceeds the decode length for that region, or if there is insufficient contiguous space to satisfy a request with Selection Policy set to Contiguous.

The command shall fail with Invalid Extent List under the following conditions:

- When the Selection Policy is set to Prescriptive and the Extent Count is invalid
- When the Selection Policy is set to Prescriptive and any of the DPAs are already accessible to the same LD

The command shall fail with Resources Exhausted if the Extent List would cause the device to exceed its extent or tag tracking ability.

The command shall fail with Retry Required if its execution would cause the specified LD's Dynamic Capacity Event Log to overflow.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required
- Invalid Extent List
- Resources Exhausted

Command Effects:

- Configuration Change after Cold Reset
- Configuration Change after Conventional Reset
- Configuration Change after CXL Reset
- Immediate Configuration Change
- Immediate Data Change

Table 7-70. Initiate Dynamic Capacity Add Request Payload

Byte Offset	Length in Bytes	Description
00h	2	Host ID: For an LD-FAM device, the LD-ID of the host interface to which the capacity is being added.
02h	1	- Bits[3:0]: Selection Policy: Specifies the policy to use for selecting which extents comprise the added capacity: 0h = Free 1h = Contiguous 2h = Prescriptive 3h = Enable Shared Access - All other encodings are reserved - Bits[7:4]: Reserved
03h	1	Region Number: Dynamic Capacity Region to which the capacity is being added. Valid range is from 0 to 7. This field is reserved when the Selection Policy is set to Prescriptive.
04h	8	Length: The number of bytes of capacity to add. Always a multiple of the configured Region Block Size returned in Get DCD Info. Shall be > 0. This field is reserved when the Selection Policy is set to Prescriptive or Enable Shared Access.
0Ch	10h	Tag: Context field utilized by implementations that make use of the Dynamic Capacity feature. This field is reserved when the Selection Policy is set to Prescriptive.
1Ch	4	Extent Count: The number of extents in the Extent List. Present only when the Selection Policy is set to Prescriptive.
20h	Varies	Extent List: Extent list of capacity to add as defined in Table 8-63. Present only when the Selection Policy is set to Prescriptive.

7.6.7.6.6 Initiate Dynamic Capacity Release (Opcode 5605h)

This command initiates the release of Dynamic Capacity for an LD-FAM DCD, from a host. This command shall complete when the device initiates the Remove Capacity procedure, as defined in Section 8.2.10.9.9. The processing of the actions initiated in response to this command may or may not result in a new entry in the Dynamic Capacity Event Log. To perform Dynamic Capacity removal on a GFD, see Section 8.2.10.9.10.8.

A removal policy is specified to govern the device's selection of which memory resources to remove:

- **Tag-based:** Extents are selected by the device based on tag, with no requirement for contiguous extents
- **Prescriptive:** Extent list of capacity to release is included in request payload

To remove a host's access to the shared extent, the FM issues Initiate Dynamic Capacity Release Request with Selection Policy=Tag-Based with the Host ID associated with that host. The Tag field must match the Tag value used during Capacity Add. The host access can be removed in any order. The physical memory resources and tag associated with a shared extent shall remain assigned and unavailable for re-use until that extent has been released from all hosts that have been granted access.

When the FM issues Initiate Dynamic Capacity Release Request with the Forced Removal flag set in order to release an extent in "Pending" state (as defined in Section 9.13.3.3), the request shall be fulfilled by the device marking the Extent Group as "Dead" without appending a new entry into the Dynamic Capacity Event Log. The Add Capacity Event records corresponding to the "Dead" Extent Group in the "Pending" list are unmodified. The "Dead" state is tracked internally by the device.

The command shall fail with Invalid Input under the following conditions:

- When the command is sent with an invalid Host ID, or an invalid region number, or an unsupported Removal Policy
- When the command is sent with a Removal Policy of Tag-based and the input Tag does not correspond to any currently allocated capacity
- When Sanitize on Release is set but is not supported by the device
- When the Tag represents sharable capacity, and the Extent List covers only a portion of the capacity associated with the Tag

The command shall fail with Resources Exhausted when the length of the removed capacity exceeds the total assigned capacity for that region or for the specified tag when the Removal Policy is set to Tag-based.

The command shall fail with Invalid Extent List when the Removal Policy is set to Prescriptive and the Extent Count is invalid or when the Extent List includes blocks that are not currently assigned to the region.

The command shall fail with Retry Required if its execution would cause the specified LD's Dynamic Capacity Event Log to overflow, unless the Forced Removal flag is set, in which case the removal occurs regardless of whether an Event is logged.

The command shall fail with Resources Exhausted if the Extent List would cause the device to exceed its extent or tag tracking ability.

The command shall fail with Invalid Physical Address if an extent in the extent list covers non-existing or pending ("Pending" state as defined in [Section 9.13.3.3](#)) DPA range and the Forced Removal flag is not set.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required
- Invalid Extent List
- Resources Exhausted

Command Effects:

- Configuration Change after Cold Reset
- Configuration Change after Conventional Reset
- Configuration Change after CXL Reset
- Immediate Configuration Change
- Immediate Data Change

Table 7-71. Initiate Dynamic Capacity Release Request Payload

Byte Offset	Length in Bytes	Description
00h	2	Host ID: For an LD-FAM device, the LD-ID of the host interface from which the capacity is being released.
02h	1	Flags - Bits[3:0]: Removal Policy: Specifies the policy to use for selecting which extents comprise the released capacity: 0h = Tag-based 1h = Prescriptive - All other encodings are reserved - Bit[4]: Forced Removal: 1 = Device does not wait for a Release Dynamic Capacity command from the host. Host immediately loses access to released capacity. - Bit[5]: Sanitize on Release: 1 = Device shall sanitize all released capacity as a result of this request using the method described in Section 8.2.10.9.5.1. If this is a shared capacity, the sanitize operation shall be performed after the last host has released the capacity. - Bits[7:6]: Reserved
03h	1	Reserved
04h	8	Length: The number of bytes of capacity to remove. Always a multiple of the configured Region Block Size returned in Get DCD Info. Shall be > 0. This field is reserved when the Removal Policy is set to Prescriptive.
0Ch	10h	Tag: Optional opaque context field utilized by implementations that make use of the Dynamic Capacity feature. This field is reserved when the Removal Policy is set to Prescriptive.
1Ch	4	Extent Count: The number of extents in the Extent List. Present only when the Removal Policy is set to Prescriptive.
20h	Varies	Extent List: Extent list of capacity to release as defined in Table 8-63 . Present only when the Removal Policy is set to Prescriptive.

7.6.7.6.7 Dynamic Capacity Add Reference (Opcode 5606h)

This command prevents the tagged sharable capacity for an LD-FAM DCD, from being sanitized, freed, and/or reallocated, regardless of whether it is currently visible to any hosts via extent lists. The tagged capacity will remain allocated, and contents will be preserved even if all DCD Extents that reference it are removed.

This command has no effect and will return Success if the FM has already added a reference to the tagged capacity.

This command shall return Invalid Input if the Tag in the payload does not match an existing sharable tag.

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- Configuration Change after Cold Reset
- Configuration Change after Conventional Reset
- Configuration Change after CXL Reset
- Immediate Configuration Change

Table 7-72. Dynamic Capacity Add Reference Request Payload

Byte Offset	Length in Bytes	Description
00h	10h	Tag: Tag that is associated with the memory capacity to be preserved.

7.6.7.6.8 Dynamic Capacity Remove Reference (Opcode 5607h)

This command removes a reference to tagged sharable capacity for an LD-FAM DCD, that was previously added via Dynamic Capacity Add Reference (see [Section 7.6.7.6.7](#)). If there are no remaining extent lists that reference the tagged capacity, the memory will be freed and sanitized if appropriate.

This command shall return Invalid Input if the Tag in the payload does not match an existing sharable tag.

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- Configuration Change after Cold Reset (if freed)
- Configuration Change after Conventional Reset (if freed)
- Configuration Change after CXL Reset (if freed)
- Immediate Configuration Change (if freed)

Table 7-73. Dynamic Capacity Remove Reference Request Payload

Byte Offset	Length in Bytes	Description
00h	10h	Tag: Tag that is associated with the memory capacity.

7.6.7.6.9 Dynamic Capacity List Tags (Opcode 5608h)

This command allows an FM to re-establish context for an LD-FAM DCD, by receiving a list of all existing tags, with bitmaps indicating which LDs have access, and a flag indicating whether the FM holds a reference.

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error

Command Effects:

- None

Table 7-74. Dynamic Capacity List Tags Request Payload

Byte Offset	Length in Bytes	Description
00h	04h	Starting Index: Index of the first tag to return.
04h	04h	Max Tags: Maximum number of tags to return in the response payload. If Max Tags is 0, no tags list will be returned; however, the Generation Number shall be valid.

Table 7-75. Dynamic Capacity List Tags Response Payload

Byte Offset	Length in Bytes	Description
00h	4	Generation Number: Generation number of the tags list. This number shall change every time the remainder of the command's payload would change.
04h	4	Total Number of Tags: Maximum number of tags to return in the response payload.
08h	4	Number of Tags Returned: Number of tags returned in the Tags List.
0Ch	1	Validity Bitmap - Bit[0]: Reference Bitmaps Valid: A value of 1 indicates that the Reference Bitmap fields in the Tags List are valid. This bit shall be 0 for GFDs and 1 for all other device types. - Bit[1]: Pending Reference Bitmaps Valid: A value of 1 indicates that the Pending Reference Bitmap fields in the Tags List are valid. This bit shall be 0 for GFDs and 1 for all other device types. - Bits[7:2]: Reserved.
0Dh	3	Reserved
10h	Varies	Tags List: List of Dynamic Capacity Tag Information structures. The format of each entry is defined in Table 7-76.

Table 7-76. Dynamic Capacity Tag Information

Byte Offset	Length in Bytes	Description
00h	10h	Tag: Tag that is associated with the memory capacity.
10h	1	Flags - Bit[0]: FM Holds Reference: When set, this bit indicates that the FM holds a reference on this Tag. - Bits[7:1]: Reserved.
11h	3	Reserved
14h	20h	Reference Bitmap: Each 1 indicates an LD that has accepted the capacity associated with this tag. Bit 0 of the first byte represents LD 0, and bit 7 of the last byte represents LD 255. This field is reserved if the Reference Bitmaps Valid bit is not set in the Dynamic Capacity List Tags Response Payload (see Table 7-75).
34h	20h	Pending Reference Bitmap: Each 1 indicates an LD for which the tagged capacity has been added with no host response yet. Bit 0 of the first byte represents LD 0, and bit 7 of the last byte represents LD 255. This field is reserved if the Pending Reference Bitmaps Valid bit is not set in the Dynamic Capacity List Tags Response Payload (see Table 7-75).

7.6.8 Fabric Management Event Records

The FM API uses the Event Records framework defined in Section 8.2.10.2.1. This section defines the format of event records specific to Fabric Management activities.

7.6.8.1 Physical Switch Event Records

Physical Switch Event Records define events that are related to physical switch ports, as defined in [Table 7-77](#).

Table 7-77. Physical Switch Events Record Format

Byte Offset	Length in Bytes	Description
00h	30h	Common Event Record: See corresponding common event record fields defined in Section 8.2.10.2.1 . The Event Record Identifier field shall be set to 77cf9271-9c02-470b-9fe4-bc7b75f2da97, which identifies a Physical Switch Event Record.
30h	1	Physical Port ID: Physical Port that is generating the event.
31h	1	Event Type: Identifies the type of event that occurred: 00h = Link State Change 01h = Slot Status Register Updated
32h	2	Slot Status Register: As defined in PCIe Base Specification.
34h	1	Reserved
35h	1	- Bits[3:0]: Current Port Configuration State: See Table 7-19 - Bits[7:4]: Reserved
36h	1	- Bits[3:0]: Connected Device Mode: See Table 7-19 - Bits[7:4]: Reserved
37h	1	Reserved
38h	1	Connected Device Type: See Table 7-19
39h	1	Supported CXL Modes: See Table 7-19
3Ah	1	- Bits[5:0]: Maximum Link Width: Value encoding matches the Maximum Link Width field in the PCIe Link Capabilities register in the PCIe Capability structure - Bits[7:6]: Reserved
3Bh	1	- Bits[5:0]: Negotiated Link Width: Value encoding matches the Negotiated Link Width field in the PCIe Link Capabilities register in the PCIe Capability structure - Bits[7:6]: Reserved
3Ch	1	- Bits[5:0]: Supported Link Speeds Vector: Value encoding matches the Supported Link Speeds Vector field in the PCIe Link Capabilities 2 register in the PCIe Capability structure - Bits[7:6]: Reserved
3Dh	1	- Bits[5:0]: Max Link Speed: Value encoding matches the Max Link Speed field in the PCIe Link Capabilities register in the PCIe Capability structure - Bits[7:6]: Reserved
3Eh	1	- Bits[5:0]: Current Link Speed: Value encoding matches the Current Link Speed field in the PCIe Link Status register in the PCIe Capability structure - Bits[7:6]: Reserved
3Fh	1	LTSSM State: See Section 7.6.7.1 .
40h	1	First Negotiated Lane Number: Lane number of the lowest lane that has negotiated.
41h	2	Link state flags: See Section 7.6.7.1 .
43h	3Dh	Reserved

7.6.8.2 Virtual CXL Switch Event Records

Virtual CXL Switch Event Records define events that are related to VCSs and vPPBs, as defined in [Table 7-78](#).

Table 7-78. Virtual CXL Switch Event Record Format

Byte Offset	Length in Bytes	Description
00h	30h	Common Event Record: See corresponding common event record fields defined in Section 8.2.10.2.1 . The Event Record Identifier field shall be set to 40d26425-3396-4c4d-a5da-3d47263af425, which identifies a Virtual Switch Event Record.
30h	1	VCS ID
31h	1	vPPB ID
32h	1	Event Type: Identifies the type of event that occurred: • 00h = Binding Change • 01h = Secondary Bus Reset • 02h = Link Control Register Updated • 03h = Slot Control Register Updated
33h	1	vPPB Binding Status: Current vPPB binding state, as defined in Table 7-32 . If Event Type is 00h, this field contains the updated binding state of a vPPB following the binding change. Successful bind and unbind operations generate events to the Informational Event Log. Failed bind and unbind operations generate events to the Warning Event Log.
34h	1	vPPB Port ID: Current vPPB bound port ID, as defined in Table 7-32 . If Event Type is 00h, this field contains the updated binding state of a vPPB following the binding change. Successful bind and unbind operations generate events to the Informational Event Log. Failed bind and unbind operations generate events to the Warning Event Log.
35h	1	vPPB LD ID: Current vPPB bound LD-ID, as defined in Table 7-32 . If Event Type is 00h, this field contains the updated binding state of a vPPB following the binding change. Successful bind and unbind operations generate events to the Informational Event Log. Failed bind and unbind operations generate events to the Warning Event Log.
36h	2	Link Control Register Value: Current Link Control register value, as defined in PCIe Base Specification.
38h	2	Slot Control Register Value: Current Slot Control register value, as defined in PCIe Base Specification.
3Ah	46h	Reserved

7.6.8.3 MLD Port Event Records

MLD Port Event Records define events that are related to switch ports connected to MLDs, as defined in [Table 7-79](#).

Table 7-79. MLD Port Event Records Payload

Byte Offset	Length in Bytes	Description
00h	30h	Common Event Record: See corresponding common event record fields defined in Section 8.2.10.2.1 . The Event Record Identifier field shall be set to 8dc44363-0c96-4710-b7bf-04bb99534c3f, which identifies an MLD Port Event Record.
30h	1	Event Type: Identifies the type of event that occurred: 00h = Error Correctable Message Received. Events of this type shall be added to the Warning Event Log. 01h = Error Non-Fatal Message Received. Events of this type shall be added to the Failure Event Log. 02h = Error Fatal Message Received. Events of this type shall be added to the Failure Event Log.
31h	1	Port ID: ID of the MLD port that is generating the event.
32h	2	Reserved
34h	8	Error Message: The first 8 bytes of the PCIe error message (ERR_COR, ERR_NONFATAL, or ERR_FATAL) that is received by the switch.
3Ch	44h	Reserved

7.7 CXL Fabric Architecture

The CXL fabric architecture adds new features to scale from a node to a rack-level interconnect to service the growing computational needs in many fields. Machine learning/AI, drug discovery, agricultural and life sciences, materials science, and climate modeling are some of the fields with significant computational demand. The computation density required to meet the demand is driving innovation in many areas, including near and in-memory computing. CXL Fabric features provide a robust path to build flexible, composable systems at rack scale that are able to capitalize on simple load/store memory semantics or Unordered I/O (UIO).

CXL fabric extensions allow for topologies of interconnected fabric switches using 12-bit PIDs (SPIDs/DPIDs) to uniquely identify up to 4096 Edge Ports. The following are the main areas of change to extend CXL as an interconnect fabric for server composability and scale-out systems:

- Expand the size of CXL fabric using Port Based Routing and 12-bit PIDs.
- Enable support for G-FAM devices (GFDs). A GFD is a highly scalable memory resource that is accessible by all hosts and all peer devices.
- Host and device peer communication may be enabled using UIO.

Figure 7-25. High-level CXL Fabric Diagram

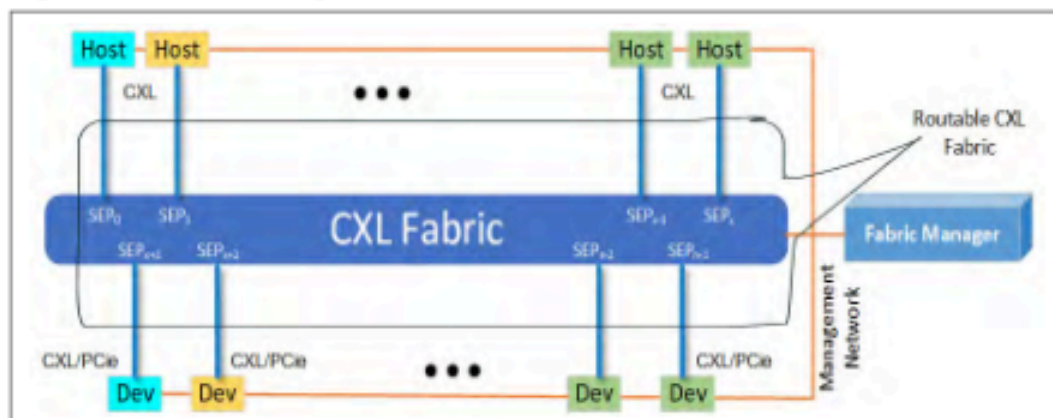


Figure 7-25 is a high-level illustration of a routable CXL Fabric. The fabric consists of one or more interconnected fabric switches. In this figure, there are “n” Switch Edge Ports (SEP) on the Fabric where each Edge Port can connect to a CXL host root port or a CXL/PCIe device (Dev). As shown, a Fabric Manager (FM) connects to the CXL Fabric and may connect to selected endpoints over an out-of-band management network. The management network may be a simple 2-wire interface, such as SMBus, I2C, I3C, or a complex fabric such as Ethernet. The FM is responsible for the initialization and setup of the CXL Fabric and the assignment of devices to different Virtual Hierarchies. Extensions to FM API (see Section 7.6) to handle cross-domain traffic will be taken up as a future ECN.

Initially, the FM binds a set of devices to the host’s Virtual Hierarchies, essentially composing a system. After the system has booted, the FM may add or remove devices from the system using fabric bind and unbind operations. These system changes are presented to the hosts by the fabric switches as managed Hot-Add and Hot-Remove events as described in Section 9.9. This allows for dynamic reconfiguration of systems that are composed of hosts and devices.

Root ports on the CXL Fabric may be part of the same or different domains. If the root ports are in different domains, hardware coherency across those root ports is not a requirement. However, devices that support sharing (including MLDs, Multi-Headed devices, and GFDs) may support hardware-managed cache coherency across root ports in multiple domains.

7.7.1 CXL Fabric Use Case Examples

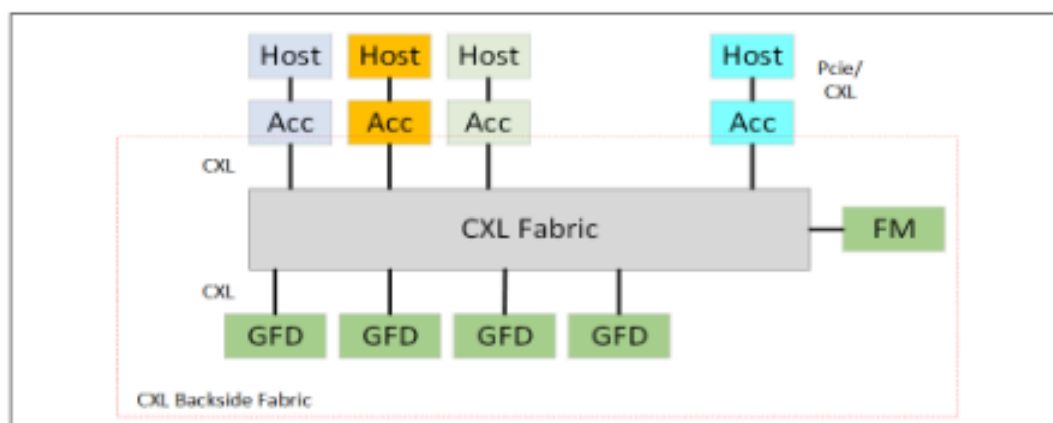
Following are a few examples of systems that may benefit from using CXL-switched Fabric for low-latency communication.

7.7.1.1 Machine-learning Accelerators

Accelerators used for machine-learning applications may use a dedicated CXL-switched Fabric for direct communication between devices in different domains. The same Fabric may also be used for sharing GFDs among accelerators. Each host and accelerator of same color shown in Figure 7-26 (basically, those that are directly above and below one another) belongs to a single domain. Accelerator devices can use UIO transactions to access memory on other accelerator and GFDs. In such a system, each accelerator is attached to a host and expected to be hardware-cache coherent with the host when using a CXL link. Communication between accelerators across domains is via the I/O coherency model. Device caching of data from another device memory (HDM or PDM) requires software-managed coherency with appropriate cache flushes and barriers. A

Switch Edge ingress port is expected to implement a common set of address decoders that is to be used for Upstream Ports and Downstream Ports. Implementations may enable a dedicated CXL Fabric for accelerators using features available in this revision. However, it is not fully defined by the specification. Peer communication is defined in [Section 7.7.9](#).

Figure 7-26. ML Accelerator Use Case

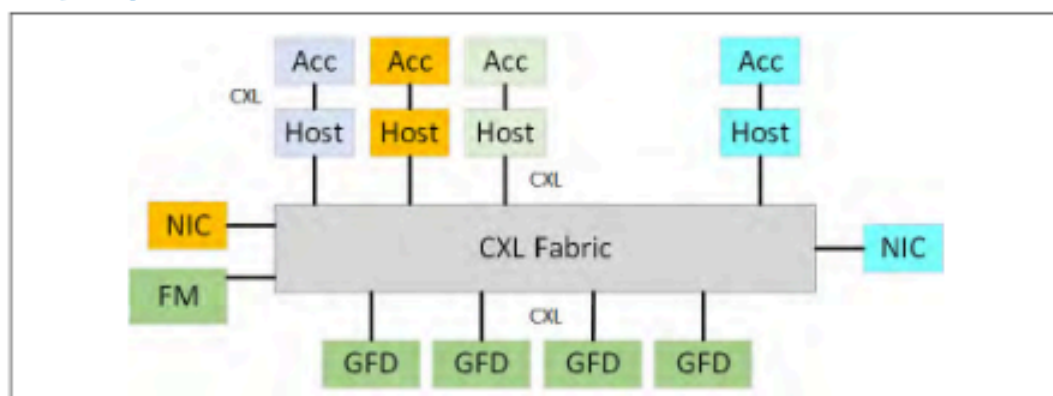


7.7.1.2 HPC/Analytics Use Case

High-performance computing and Big Data Analytics are two areas that may also benefit from a dedicated CXL Fabric for host-to-host communication and sharing of G-FAM. CXL.mem or UIO may be used to access GFDs. Some G-FAM implementations may enable cross-domain hardware cache coherency. Software cache coherency may still be used for shared-memory implementations. Host-to-host communication is defined in [Section 7.7.3](#).

NICs may be used to directly move data from network storage to G-FAM devices, using the UIO traffic class. CXL.mem and UIO use fabric address decoders to route to target GFDs that are members of many domains.

Figure 7-27. HPC/Analytics Use Case

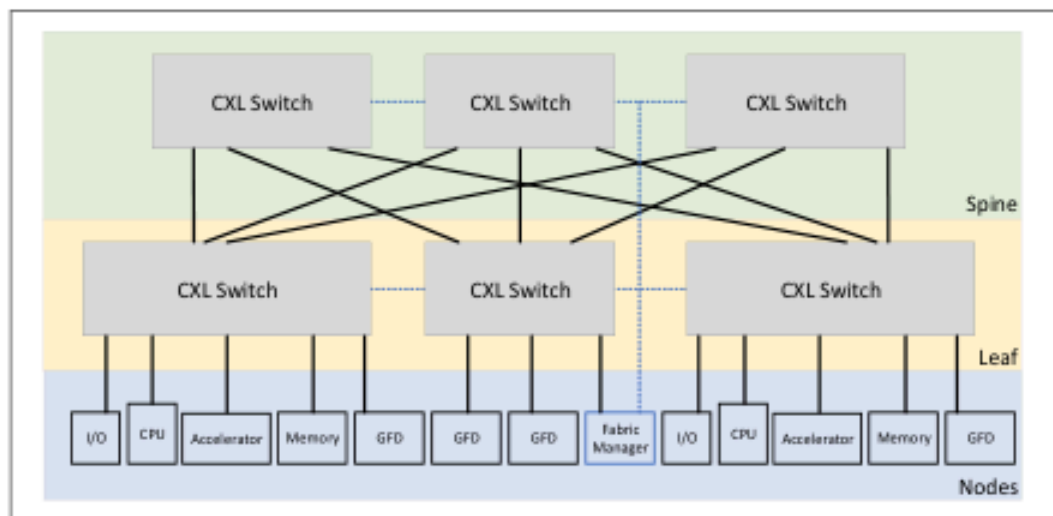


7.7.1.3 Composable Systems

Support for multi-level switches with PBR fabric extensions provides additional capabilities for building software-composable systems. In [Figure 7-28](#), a leaf/spine switch architecture is shown in which all resources are attached to the leaf switches. Each domain may span multiple switches. All devices must be bound to a host or an FM. Cross-domain traffic is limited to CXL.mem and UIO transactions.

Composing systems from resources within a single leaf switch allows for low-latency implementations. In such implementations, a spine switch is used only for cross-domain and G-FAM accesses.

Figure 7-28. Sample System Topology for Composable Systems



7.7.2 Global-Fabric-Attached Memory (G-FAM)

7.7.2.1 Overview

G-FAM provides a highly scalable memory resource that is accessible by all hosts and peer devices within a CXL fabric. G-FAM ranges can be assigned exclusively to a single host/peer requester or can be shared by multiple hosts/peers. When shared, multi-requester cache coherency can be managed by either software or hardware. Access rights to G-FAM ranges are enforced by decoders in Requester Edge ports and the target GFD.

GFD HDM space can be accessed by hosts/peers from multiple domains using CXL.mem, and by peer devices from multiple domains using CXL.io UIO. GFDs implement no PCIe configuration space, and they are configured and managed instead via Global Memory Access Endpoints (GAEs) in Edge USPs or via out-of-band mechanisms.

Unlike an MLD, which has a separate Device Physical Address (DPA) space for each host/peer interface (LD), a GFD has one DPA space that is common across all hosts and peer devices. The GFD translates the Host Physical Address (HPA)¹ in each incoming request into a DPA, using per-requester translation information that is stored within the GFD Decoder Table. To create shared memory, two or more HPA ranges (each from a different requester) are mapped to the same DPA range. When the GFD needs to issue a BISnp, the GFD translates the DPA into an HPA for the associated host using the same GFD decoder information.

When a GFD receives a request, the requester is identified by the SPID in the request, which is referred to as the Requester PID or RPID. Using this term avoids confusion when describing messages that the GFD sends to the requester, where the RPID is used for the DPID, and the GFD PID is used for the SPID.

1. "HPA" is used for peer device requests in addition to host requests, even though "HPA" is a misnomer for some peer-device use cases.

All memory capacity on a GFD is managed by the Dynamic Capacity (DC) mechanisms, as defined in [Section 8.2.10.9.9](#). A GFD allows each requester to access up to 8 RPID non-overlapping decoders, where the maximum number of decoders per SPID is implementation dependent. Each decoder has a translation from HPA space to the common DPA space, a flag that indicates whether cache coherency is maintained by software or hardware, and information about multi-GFD interleaving, if used. For each requester, the FM may define DC Regions in DPA space and convey this information to the host via a GAE. It is expected that the host will program the Fabric Address Segment Table (FAST) decoders and GFD decoders for all RPIDs in its domain to map the entire DPA range of each DC Region that needs to be accessed by the host or by one of its associated accelerators.

G-FAM memory ranges can be interleaved across any power-of-two number of GFDs from 2 to 256, with an Interleave Granularity of 256B, 512B, 1 KB, 2 KB, 4 KB, 8 KB, or 16 KB. GFDs that are located anywhere within the CXL fabric, as defined in [Section 2.7](#), may be used to contribute memory to an Interleave Set.

If a GFD supports UIO Direct P2P to HDM (see [Section 7.7.9.1](#)), all GFD ports shall support UIO, and for each GFD link whose link partner also supports UIO, VC3 shall be auto-enabled by the ports (see [Section 7.7.11.5.1](#)).

7.7.2.2 Host Physical Address View

Hosts that access G-FAM shall allocate a contiguous address range for Fabric Address space within their Host Physical Address (HPA) space, as shown in [Figure 7-29](#). The Fabric Address range is defined by the FabricBase and FabricLimit registers. All host requests that fall within the Fabric Address range are routed to a selected CXL port. Hosts that use multiple CXL ports for G-FAM may either address interleave requests across the ports or may allocate a Fabric Address space for each port.

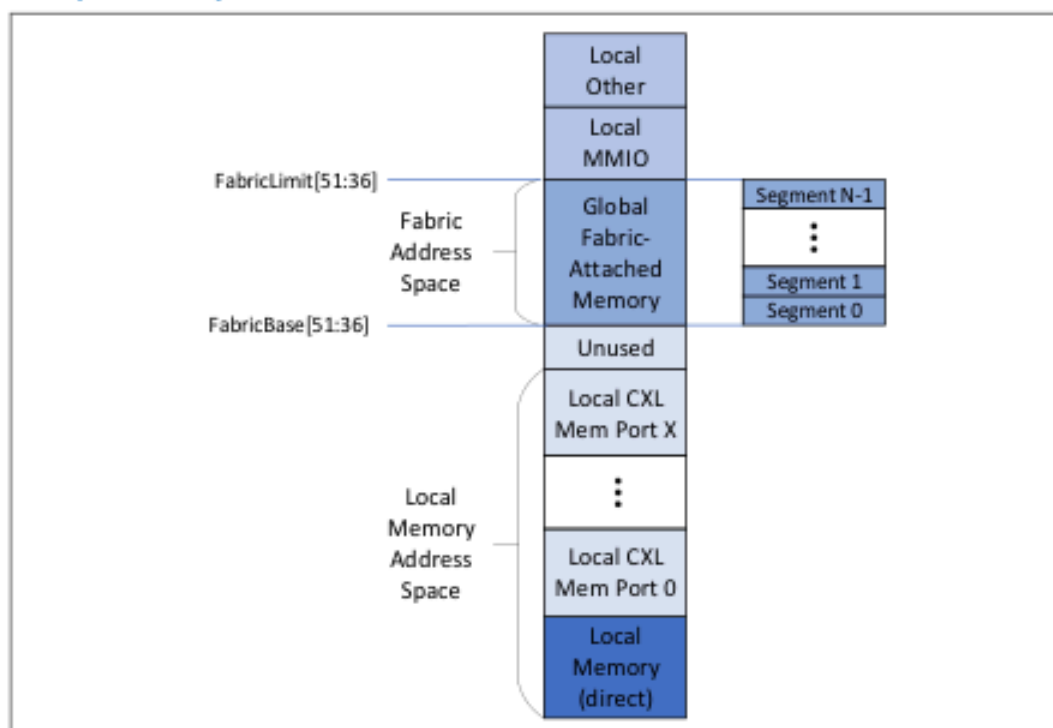
G-FAM requests from a host flow to a PBR Edge USP. In the USP, the Fabric Address range is divided into N equal-sized segments. A segment may be any power-of-two size from 64 GB to 8 TB, and must be naturally aligned. The number of segments implemented by a switch is implementation dependent. Host software is responsible for configuring the segment size so that the number of segments times the segment size fully spans the Fabric Address space. The FabricBase and FabricLimit registers can be programmed to any multiple of the segment size.

Each segment has an associated GFD or Interleave Set of GFDs. Requests whose HPA falls anywhere within the segment are routed to the specified GFD or to a GFD within the Interleave Set. Segments are used only for request routing and may be larger than the accessible portion of a GFD. When this occurs, the accessible portion of the GFD starts at address offset zero within the segment. Any requests within the segment that are above the accessible portion of the GFD will fail to positively decode in the GFD and will be handled as described in [Section 8.2.4.20](#).

Host interleaving across root ports is entirely independent from GFD interleaving. Address bits that are used for root port interleaving and for GFD interleaving may be fully overlapping, partially overlapping, or non-overlapping. When the host uses root port interleaving, FabricBase, FabricLimit, and segment size in the corresponding PBR Edge USPs must be identically configured.

7.7.2.3 G-FAM Capacity Management

Figure 7-29. Example Host Physical Address View



GFDs are managed using CCIs like all other classes of CXL components. A GFD requires support for the PBR Link CCI message format, as defined in [Section 7.7.11.6](#), on its CXL link and may optionally implement additional MCTP-based CCIs (e.g., SMBus).

G-FAM relies exclusively on the Dynamic Capacity (DC) mechanism for capacity management, as described in [Section 8.2.10.9.9](#). GFDs have no “legacy” static capacity as shown in the left side of [Figure 9-24](#) in [Chapter 9.0](#). Dynamic Capacity for G-FAM has much in common with the Dynamic Capacity for LD-FAM:

- Both have identical concepts for DC Regions, Extents, and Blocks
- Both support up to 8 DC Regions per host/peer interface
- DC-related parameters in the CDAT for each are identical
- Mailbox commands for each are highly similar; however, the specific Mailbox access methods are considerably different
 - For LD-FAM, the Mailbox for each host’s LD is accessed via LD structures
 - For G-FAM, management for each host is defined in [Section 7.7.2.6](#)

An LD-FAM DCD (i.e., DCD-capable SLDs or MLDs) allocates memory capacity and binds it to a specific Host ID in one operation. A GFD allocates Dynamic Capacity to a named Memory Group in one operation and binds specific Host IDs to named Memory Groups in a separate operation. Thus, the GFD requires different DCD Management commands than LD-FAM DCDs.

In contrast to LD-FAM, each GFD has a single DPA space instead of a separate DPA space per host. G-FAM DPA space is organized by Device Media Partitions (DMPs), as shown in [Figure 7-30](#). DMPs are DPA ranges with certain attributes. A fundamental DMP

attribute is the media type (e.g., DRAM or PM). A DMP attribute that is configured by the FM is the DC Block size. DMPs expose all GFD memory that is assignable for host use.

The rules for DMPs are as follows:

- Each GFD contains 1–4 DMPs, whose size is configured by the FM.
- Each DC Region consists of part or all of one DMP assigned to a host/peer. Each DC Region can be mapped into an RPID's HPA space using the GFD Decoder Table.
- Each DC Region inherits associated DMP attributes.

Figure 7-30. Example HPA Mapping to DMPs

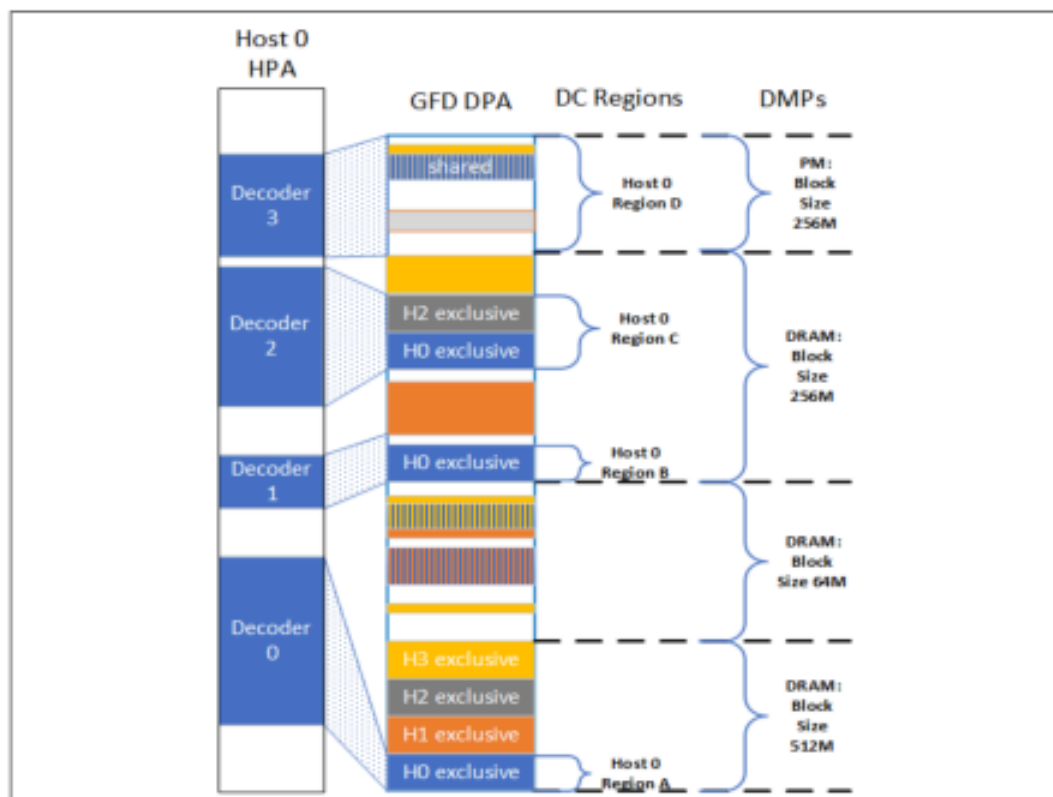


Table 7-80 lists the key differences between LD-FAM and G-FAM.

Table 7-80. Differences between LD-FAM and G-FAM (Sheet 1 of 2)

Feature or Attribute	LD-FAM	G-FAM
Number of supported hosts	16 max	1000s architecturally; 100s more realistic
Support for DMPs	No	Yes
Architected FM API support for DMP configuration by the FM	N/A	Yes

Table 7-80. Differences between LD-FAM and G-FAM (Sheet 2 of 2)

Feature or Attribute	LD-FAM	G-FAM
Routing and decoders used for HDM addresses	Interleave RP routing by host HDM Decoder Interleave VH routing by USP HDM Decoder Interleave fabric routing by USP LDST/IDT decoder 1–10 HDM Decoders in each LD	Interleave RP routing by host HDM Decoder Interleave fabric routing by USP FAST/IDT decoder 1–8 GFD Decoders per RPID in the GFD
Interleave Ways (IW)	1/2/4/8/16 plus 3/6/12	2–256 in powers of 2
DC Block Size	Powers of 2, as indicated by Region * Supported Block Size Mask	64 MB and up in powers of 2

Additional differences exist in how MLDs and GFDs process requests. An MLD has three types of decoders that operate sequentially on incoming requests:

- Per-LD HDM decoders translate from HPA space to a per-LD DPA space, removing the interleaving bits
- Per-LD decoders determine within which per-LD DC Region the DPA resides, and then whether the addressed DC block within the Region is accessible by the LD
- Per-LD implementation-dependent decoders translate from the DPA to the media address

A GFD has two types of decoders that operate sequentially on incoming requests:

- Per-RPID GFD decoders translate from HPA space to a common DPA space, removing the interleaving bits. This DPA may be used as the media address directly or via a simple mapping.
- A common decoder determines within which Device Media Partition (DMP) the DPA is located, and then whether the block that is addressed within the DMP is accessible by the RPID.

7.7.2.4 G-FAM Request Routing, Interleaving, and Address Translations

The mechanisms for GFD request routing, interleaving, and address translations within both the Edge ingress port and the GFD are shown in [Figure 7-31](#). GFD requests may arrive either at an Edge USP from a host or at an Edge DSP from a peer device. This is referred to as the Edge request port.